

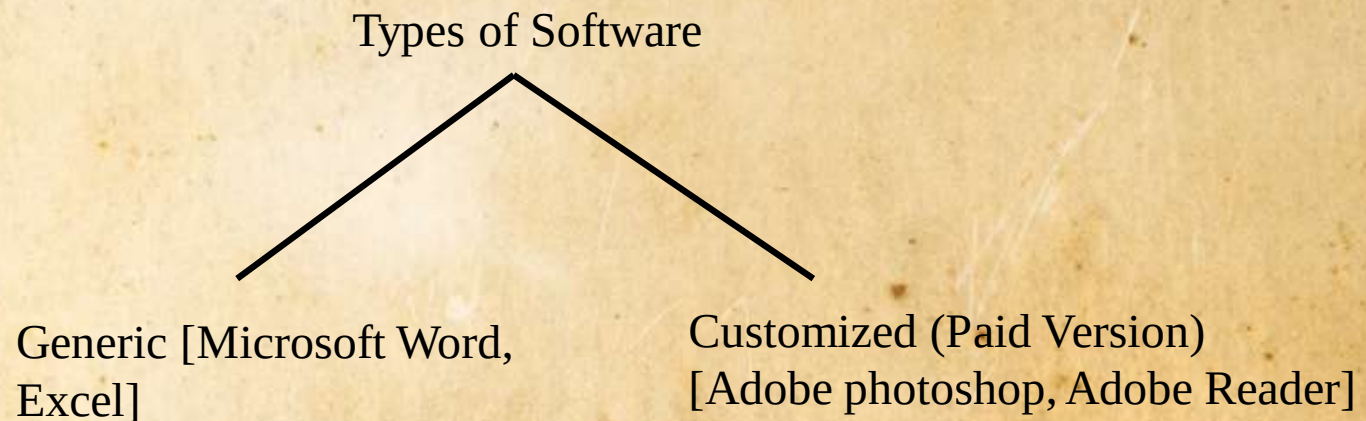
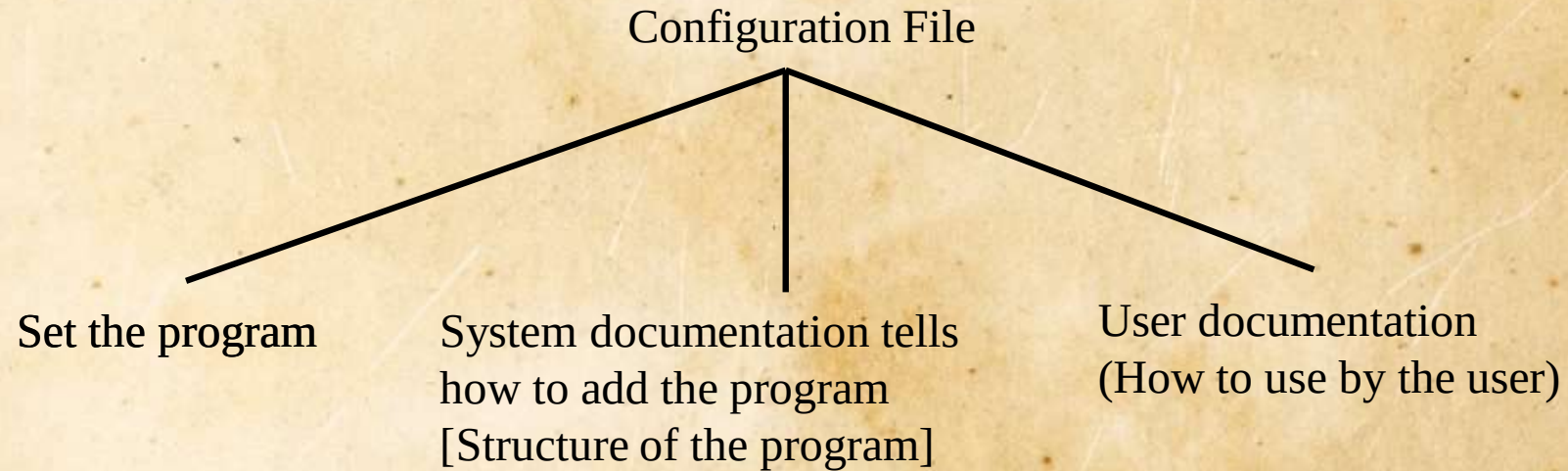
CSE-327 (Software Engineering)
Faculty: AKM Iqtidar Newaz (IqN)
Lecture - 1

What is a Software?

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A set of programs. Initially, total works are divided into different parts and before releasing merged every parts all together to deliver the software.

What can be there in Software?



What is a Software Engineering?

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Software engineering is the application of a systematic, disciplined, cost-effective techniques, which is an engineering approach for the development, operation and maintenance of a software.



Difference between Software Engineering and Computer Science?

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Computer science deals with the science behind interaction between hardware and software systems and computational applications.

Essential Attributes of Good Software



- ☐ Functionality - fulfilling the goal and requirement
- ☐ Usability - easy and flexible to use.
- ☐ Efficiency - no extra memory space
 - responsiveness
 - fast processing time.
- ☐ Maintainability - easy to change the design or add new feature.
- ☐ Security - no unauthorized access.
- ☐ Reliability - if software fails, there must be a backup system.

What is a Software Process?

- ☐ Software specification - users and developers meet, explain their expectations, budgets (SRS[Software Requirement Specification]).
- ☐ Software development - design and program.
- ☐ Software validation - check with the SRS.
- ☐ Software evaluation - if any changes need, we can change it.



What are the key challenges in Software Engineering?

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- ❑ Legacy challenge – Most softwares were created long time ago. It is being updated over times by developers.

Updating this software is a big challenge and this is called legacy challenge.

- ❑ Heterogeneity challenge – Every day OS updates, network updates, all these updates can work smoothly.
- ❑ Delivery challenge – Small time limit but need to deliver high quality software.

What is SDLC?

What is SDLC?

- ❑ SDLC stands for Software Development Life Cycle. Also referred to as the Application Development Life Cycle.
- ❑ Systemic process to develop a software.
- ❑ SDLC process aims to produce high-quality software that meets customer expectations.

What is SDLC?



- ❑ The system development should be complete within the pre-defined time frame and cost.
- ❑ SDLC consists of a detailed plan which explains how to plan, build, and maintain specific software.
- ❑ Every phase of the SDLC life cycle has its own process and deliverables that feed into the next phase.

Why SDLC?

- ☐ It provides basis for project planning, scheduling and estimating.
- ☐ Provides a framework for a standard set of activities and deliverables.
- ☐ It is a mechanism for project tracking and control.
- ☐ Increases visibility of project planning to all involved stakeholders of the development process.
- ☐ Increased and enhance development speed.
- ☐ Improved client relations.
- ☐ Helps to decrease project risk and project management plan overhead.